

CHENGDU SHUANGLIU, China

WMO: 562940

Lat: 30.579N

Lon: 103.947E

Elev: 495

StdP: 95.51

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	0.9	2.0	-5.2	2.6	8.3	-2.9	3.1	7.3	5.9	6.8	5.1	7.0	0.8	350	0.322

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.5	34.2	24.9	33.1	24.5	32.0	24.0	27.1	31.3	26.3	30.3	25.7	29.4	2.0	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.1	22.8	29.3	25.2	21.6	28.6	24.8	21.1	28.3	89.3	31.4	85.9	30.8	83.1	29.7	31.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 5.8	4.6	3.9	-1.4	36.5	1.0	1.2	-2.1	37.3	-2.7	38.0	-3.3	38.7	-4.1	39.6		
(5)			-1.8	28.4	1.1	1.1	-2.6	29.1	-3.2	29.7	-3.8	30.3	-4.6	31.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	17.8	6.7	9.3	13.6	18.7	22.5	25.0	26.9	26.7	22.9	18.5	13.6	8.4
	DBStd	7.48	2.31	3.01	3.56	3.52	2.59	2.40	2.11	2.50	2.71	2.55	2.88	2.23
	HDD10.0	220	103	44	8	1	0	0	0	0	0	0	4	60
	HDD18.3	1284	359	252	153	37	2	0	0	0	1	28	142	309
	CDD10.0	3059	2	25	118	262	389	449	523	516	388	264	114	9
	CDD18.3	1081	0	0	5	48	133	199	264	258	139	34	1	0
	CDH23.3	8720	0	0	37	344	917	1577	2521	2429	794	101	1	0
	CDH26.7	3106	0	0	3	89	287	532	969	977	233	15	0	0
Wind	WSAvg	1.5	1.3	1.5	1.7	1.7	1.8	1.7	1.6	1.6	1.6	1.4	1.3	1.2
Precipitation	PrecAvg	887	9	12	24	51	74	116	216	205	116	41	16	7
	PrecMax	1365	21	30	52	133	173	234	552	379	244	77	37	17
	PrecMin	641	0	1	3	9	20	29	76	84	34	16	3	1
	PrecStd	164	6	9	12	23	34	54	105	76	60	18	8	4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	16.2	20.2	26.9	31.3	33.8	34.2	35.9	36.2	33.8	28.8	23.0	16.2
		MCWB	9.3	12.3	16.5	19.5	20.6	22.9	25.8	25.2	24.7	19.9	16.2	10.3
	2%	DB	13.8	18.0	24.0	29.1	31.2	32.8	34.1	34.2	31.2	26.0	21.0	14.2
		MCWB	8.4	11.4	15.2	18.9	19.9	22.8	25.4	25.6	23.5	19.2	15.5	9.5
	5%	DB	12.0	16.0	21.9	27.1	29.9	31.1	33.0	33.1	29.2	24.0	19.2	13.1
		MCWB	7.8	10.5	14.5	18.5	20.1	22.7	25.2	25.4	22.8	18.5	14.6	9.1
	10%	DB	10.7	14.1	19.8	25.1	28.1	29.9	31.5	31.8	27.2	22.2	17.9	12.0
		MCWB	7.1	9.9	13.8	17.7	19.5	22.8	25.0	25.0	22.1	18.2	14.3	8.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.2	13.1	18.2	22.2	23.6	26.7	28.5	28.0	26.5	22.3	17.3	12.0
		MCDB	14.0	18.4	24.0	28.5	28.7	30.2	32.7	32.4	31.2	25.5	21.1	14.3
	2%	WB	9.2	12.1	16.4	20.5	22.4	25.5	27.3	27.0	25.0	20.6	16.4	10.9
		MCDB	12.4	16.3	21.3	25.8	27.0	28.9	31.2	31.7	28.8	23.5	19.4	13.1
	5%	WB	8.4	11.2	15.2	19.5	21.6	24.7	26.5	26.3	23.9	19.7	15.7	10.1
		MCDB	11.1	15.2	20.0	24.9	26.1	28.0	30.2	30.6	27.6	22.5	18.1	12.1
	10%	WB	7.6	10.2	14.2	18.4	20.9	23.9	25.9	25.7	22.9	18.9	14.9	9.3
		MCDB	9.9	13.6	18.8	23.7	25.7	27.4	29.3	29.7	26.3	21.5	17.0	11.2
Mean Daily Temperature Range	5% DB	MDBR	5.8	6.5	7.7	8.7	8.6	7.5	7.5	7.8	6.4	5.6	5.7	5.6
		MCDBR	9.1	10.2	11.8	12.5	12.3	10.4	10.4	10.2	9.7	8.9	8.4	8.1
		MCWBR	4.9	4.9	5.0	4.5	3.9	3.6	3.9	3.7	3.7	3.5	3.9	4.4
	5% WB	MCDBR	8.1	9.2	10.3	10.6	9.0	8.0	8.6	8.8	8.3	7.1	6.9	6.5
		MCWBR	4.8	4.7	4.8	4.6	3.8	4.0	4.1	3.9	3.9	3.4	3.6	3.9
Clear-Sky Solar Irradiance	taub	0.680	0.722	0.756	0.743	0.704	0.673	0.700	0.730	0.746	0.753	0.733	0.690	
	taud	1.475	1.432	1.415	1.476	1.585	1.672	1.633	1.564	1.516	1.471	1.460	1.492	
	Ebn at Noon	529	561	587	624	654	673	652	624	587	533	487	493	
	Edh at Noon	252	287	310	301	271	248	257	272	276	271	254	238	
All-Sky Solar Radiation	RadAvg	1.97	2.40	3.13	3.94	4.23	3.97	4.25	4.09	2.92	2.32	2.03	1.86	
	RadStd	0.26	0.36	0.34	0.40	0.49	0.49	0.60	0.56	0.35	0.32	0.16	0.27	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		+0.56	+0.50	-1.23	+0.99	N/A	N/A	-50	-91	+164
(47) Regional (6 neighbors)		+0.28	N/A	-0.64	+0.41	N/A	N/A	-50	-91	+164

Nomenclature: See separate page